

Maleck Affie

Senior Data Architect | Cloud Data Platforms & Lakehouse Architecture |
Spark, Kafka, Databricks | AWS, Azure, GCP

✉ maleckaffie20@gmail.com ☎ (908) 639-3626 📍 Warren, NJ

SUMMARY

Result-driven Senior Data Architect with 10+ years of experience designing enterprise-scale, cloud-native data platforms and lakehouse architectures. Expert in real-time streaming, batch/ELT pipelines, and AI/LLM-ready analytics environments across AWS, Azure, and GCP. Proven track record in modernizing legacy systems, enabling data mesh strategies, and implementing governance frameworks (Collibra, Alation) to ensure GDPR, HIPAA, SOC 2 compliance. Skilled at orchestrating multi-cloud solutions, integrating vector databases, and delivering low-latency insights for enterprise decision-making.

SKILLS & TECHNOLOGIES

- **Data Architecture & Governance:** Lakehouse Architecture, Data Modeling (Star, Snowflake, Data Vault), Data Warehousing, Master Data Management, Data Lineage, Data Quality, Metadata Management, Data Mesh, Data Governance (Collibra, Alation, GDPR, HIPAA, SOC 2)
- **Big Data & Processing:** Apache Spark, PySpark, Scala, Apache Flink, Apache Beam, Databricks, Delta Lake, Apache Iceberg, Hadoop, Hive
- **Streaming & Real-Time:** Apache Kafka, Kafka Streams, AWS Kinesis, Google Pub/Sub, RabbitMQ
- **Cloud Platforms:** AWS (S3, Glue, Lambda, Redshift, Athena), Azure (Data Lake, Synapse, Data Factory), GCP (BigQuery, Cloud Storage), Snowflake
- **Databases & Warehouses:** Snowflake, Amazon Redshift, BigQuery, PostgreSQL, MySQL, MongoDB, DynamoDB, Cassandra, SQL Server, Oracle
- **Pipelines & Orchestration:** Apache Airflow, dbt, AWS Glue, Azure Data Factory, Informatica, Talend, Apache NiFi
- **AI/ML & LLM Integration:** LangChain, Pinecone, FAISS, OpenAI APIs, Hugging Face, TensorFlow, PyTorch, MLflow, MLOps, Feature Engineering
- **DevOps & IaC:** Terraform, Docker, Kubernetes, Jenkins, Git, CI/CD, AWS CloudFormation
- **Programming & Analytics:** Python, SQL, Java, Shell Scripting, Tableau, Power BI, Looker, Prometheus, Grafana, DataDog
- **Methodologies:** Agile, Scrum, Kanban, Remote Leadership, Stakeholder Management, Full SDLC

Professional Experience

Data Architect

12/2022 – Present

Datafold

- Designed and led Databricks Lakehouse platform supporting multiple data domains and pipelines, enabling real-time analytics and AI/LLM workflows.

- Architected high-throughput streaming pipelines using Apache Flink and Kafka, supporting critical business decisions with low-latency data.
- Modernized legacy ETL pipelines into cloud-native ELT frameworks using dbt, AWS Lambda, Kubernetes, and Airflow, improving data delivery speed by 30%.
- Designed and optimized scalable database solutions using PostgreSQL, MongoDB, and DynamoDB for high-volume data ingestion and analytics workloads.
- Integrated LLM-powered analytics with LangChain, Pinecone, and FAISS, enabling natural language queries and AI-driven insights for stakeholders.
- Established enterprise data governance framework, enforcing data lineage, quality standards, and compliance with GDPR, HIPAA, and SOC 2.

Senior Data Engineer

06/2017 – 11/2022

Numerai

- Developed batch and real-time ETL/ELT pipelines using Apache Spark, Spark Streaming, Kafka, Python, and Airflow, reducing processing time by ~25%.
- Designed and optimized Snowflake, Amazon Redshift, and BigQuery data warehouse environments, improving query performance and enabling scalable analytics for BI and data science teams.
- Implemented automated data quality and monitoring frameworks, ensuring reliable datasets for ML and analytics teams.
- Led multi-cloud migration projects across AWS, Azure, and GCP, improving scalability and cost efficiency.
- Developed feature engineering pipelines for ML initiatives, accelerating model training and deployment.

Data Engineer

12/2015 – 05/2017

Trifacta

- Built scalable data ingestion pipelines for structured and unstructured sources using Python, SQL, Spark, and Airflow.
- Designed dimensional data models (star and snowflake schemas) supporting analytics and reporting.
- Migrated ETL workflows to AWS cloud infrastructure, improving pipeline reliability by 20%.
- Optimized pipeline performance to minimize downtime and failures, ensuring timely dataset delivery.
- Collaborated with analytics teams to deliver high-quality datasets supporting product decisions.

Junior Data Engineer

10/2014 – 11/2015

SparkBeyond

- Maintained SQL-based ETL workflows for enterprise reporting systems with consistent data accuracy.
- Supported Hadoop-based batch processing (Hive/HDFS) for large-scale data handling.
- Optimized complex SQL queries to improve execution performance.

- Automated repetitive data processing tasks using Python, increasing efficiency.
- Assisted in managing relational databases including PostgreSQL and MySQL, supporting data extraction, transformation, and reporting for analytics teams.

Education

Bachelor's Degree in Computer Science

Preston Royal Institute of Education